```
## Error in `prettyNum()`:
## ! invalid value 0 for 'digits' argument
```

An Automatic Finite-Sample Robustness Metric: Can Dropping a Little Data Make a Big Difference?

Ryan Giordano (rgiordano@berkeley.edu)[1]

JSM 2026: Data Selection Beyond Estimation Efficiency

Dropping data: Mexico Microcredit

Example: Angelucci et al. [2015], a randomized controlled trial study of the efficacy of microcredit in Mexico based on $N =$16,560 data points. A regression was run to estimate the average effect of microcredit.

Original result: Treatment effect statistically insignificant at 95%.

Policy implication: Disinvest in microcredit initiatives.

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Cannot find influential subsets by brute force!

We provide a fast, automatic tool to approximately identify the most influential set of points.

Outline

- How do we identify influential sets? When is our method accurate?
- Data dropping as distributional robustness
- Do we identify "the most important datapoints?"

Example: Least squares

Formalizing the question.

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A data point d_n has regressors x_n and response y_n: $d_n = (x_n, y_n)$.

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$$\Leftrightarrow \sum_{n=1}^{N} \left(y_n - \hat{\theta}^T x_n\right) x_n = 0.$$

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- A particular component: $\hat{\theta}_k$
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Make a qualitative decision using $\phi(\hat{\theta})$ for a smooth, real-valued ϕ.

(WLOG try to increase $\phi(\hat{\theta})$.)

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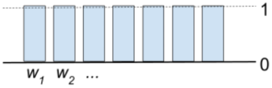
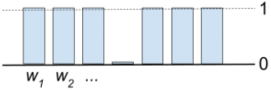


Leave points out by setting their elements of w to zero.



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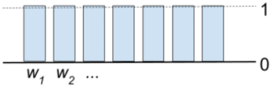

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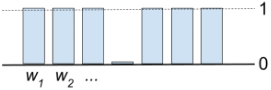
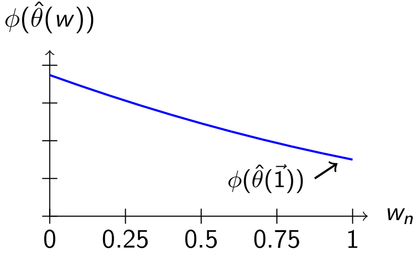


The map $w \mapsto \phi(\hat{\theta}(w))$ is well-defined even for continuous weights.

Taylor series approximation.



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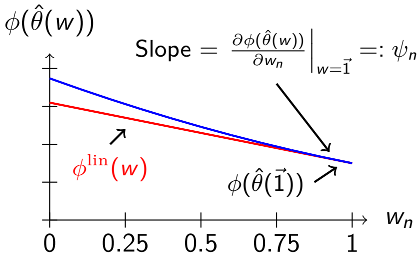

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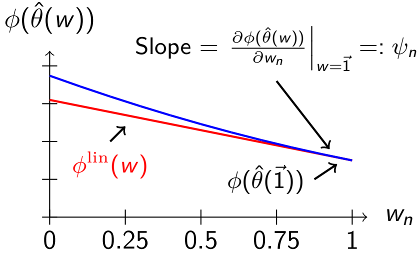

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We call ψ_n an "influence score."

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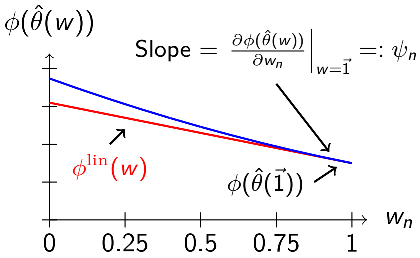

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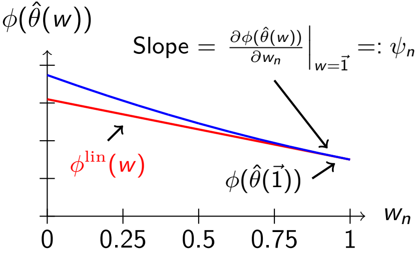

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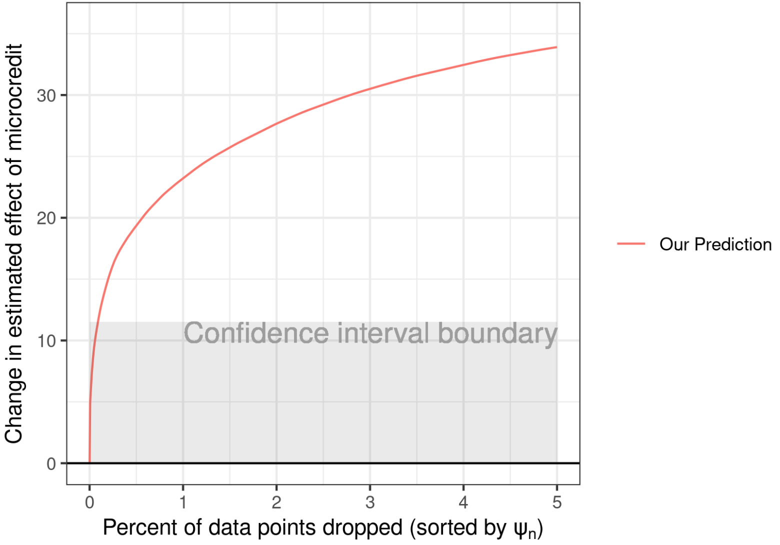
The most influential points to drop from the linear approximation are easy to identify — they are simply the points with the largest ψ_n!

How accurate is the approximation?

Checking the approximation for Mexico microcredit.



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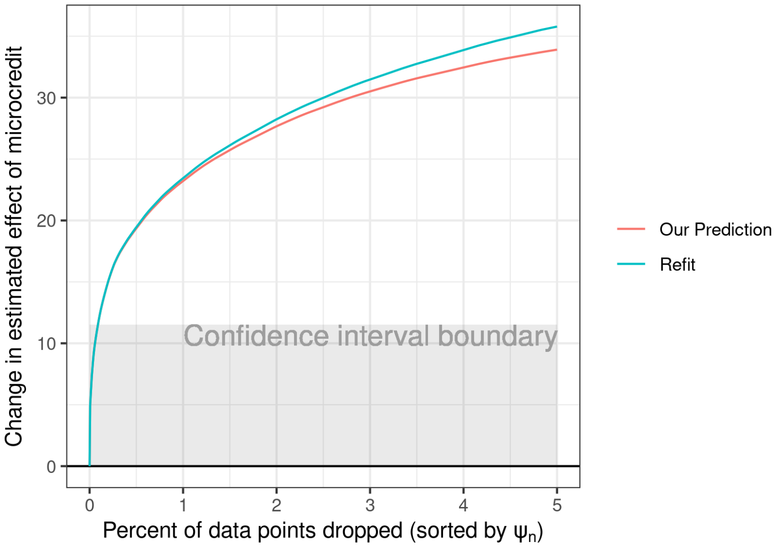
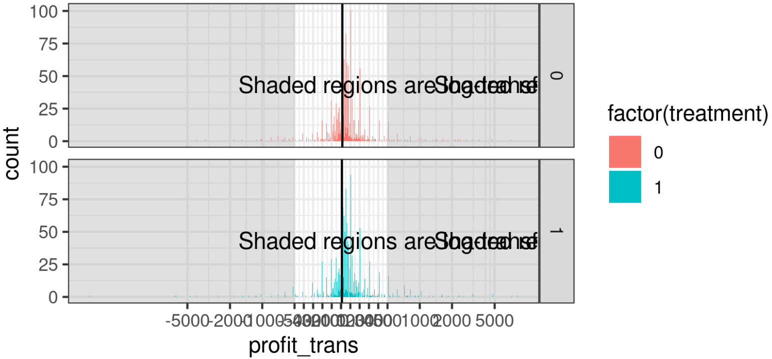


Distributional robustness



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- We can quickly and automatically find an approximate influential set which is accurate for small sets.
- Data dropping robustness is principally determined by the signal to noise ratio, and captures sensitivity distinct from sampling and gross error sensitivity.

Links and references

Tamara Broderick, Ryan Giordano, Rachael Meager (alphabetical authors)
"An Automatic Finite-Sample Robustness Metric: Can Dropping a Little Data Change Conclusions?"
https://arxiv.org/abs/2011.14999

Select blog posts with more details: `https://rgiordan.github.io`

- Data dropping sensitivity overcomes p-hacking
- Collinearity in OLS after dropping
- Influence functions and sums
- Connections to the bootstrap

Related software on `github`:

- rgiordan/zaminfluence (for `R`)
- rgiordan/vittles (for `Python`)

Some of my work on other forms of robustness:

- Prior sensitivity in Bayesian nonparametrics [Giordano et al., 2021]
- Approximate cross-validation (and other reweightings) [Giordano et al., 2019b,a]
- Covariances and prior sensitivity for mean field VB [Giordano et al., 2015, 2018]
- Model sensitivity of MCMC output [Giordano et al., 2018]
- Frequentist variances of MCMC posteriors (in progress)

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